

REVIEW

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Diabetic foot prevention, assessment, and management using innovative smart wearable technology: a systematic review

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Abstract

The diabetic foot (DF) is a chief culprit behind significant preventable morbidity and mortality in type-2-diabetes patients and the leading cause of non-traumatic lower-extremity amputations. Despite the clinically well-understood pathways to ulceration, including neuropathy, ischemia, and infection, DF continues to impose a significant health and economic burden on patients and healthcare systems. Recent technological developments in sensing, smart miniaturized wearable technology, and artificial intelligence provide ample options for viable solutions. A continuous monitoring wearable system that alerts patients and healthcare providers could enhance current detection and management protocols. Quantification of both mechanical (plantar-pressure and shear-forces) and physiological/wound-associated parameters (temperature, humidity, vascular, microcirculation, and pH) underlying the pathophysiology of DF provides opportunities for improving the assessment of DF and treatment outcomes. This review bridges this knowledge gap by exploring the integration of current state-of-the-art sensing modalities and smart wearables with novel actuation technology for real-time management towards effective wound healing of the DF.

Keywords Diabetic foot, Diabetic neuropathy, Peripheral vascular diseases (PVD), Sensors

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Introduction

Type 2 Diabetes (T2D) is currently considered the fastest-growing disease in the world [1]. Complications affecting the lower limbs are considered the most common debilitating manifestations of T2D, accounting for significant preventable morbidity and mortality by predisposing the foot to neuropathy, infections, and ulcerations, which could lead to lower extremity amputation. It is estimated that around 20% of patients with DF (1 in every 5 patients) require amputation [2], rendering the Diabetic Foot (DF) as the primary cause of non-traumatic lower extremity limb loss, worldwide (> 85% of global amputation) [3]. Importantly, once ulcerated, DF ulcer recurrence rates are 40% within the first year and increase to 65% after 5 years [4]. The economic implications are



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equally important where global healthcare expenditures associated with T2D surpassed \$966 Billion in 2021 [5].

DF has a multifactorial pathology with at least three main etiological factors, i.e., peripheral neuropathy, peripheral arterial disease, and infection [6]. Independent baseline predictors of non-healing DF ulcers include old age, gender (male sex), heart failure, inability to stand or walk without help, end-stage renal disease, larger ulcer size, diabetic peripheral neuropathy (DPN), and peripheral arterial disease (PAD). Infection has also emerged as a specific predictor of non-healing, although mainly in PAD patients [6].

DPN affects around 60% of diabetic patients due prolonged hyperglycemia which impairs blood vessel elasticity and oxygen delivery to the extremities [7]. The resulting neural ischemia disrupts neural signal transmission, leading to symptoms such as numbness, paresthesia, and sensory loss. Poor sweat gland function in the foot causes extreme skin dryness, among other issues [7]. Additionally, persistent pressure and minor trauma to the feet can lead to callus formation. Without monitoring and prompt care, these calluses may deteriorate into ulcers. DF ulcers (DFU) are typically classified into three main types: neuropathic (wound mainly due to peripheral neuropathy), ischemic (wound due to ischemia), and neuro-ischemic (wound due to both neuropathy and ischemia) [7].

The intricate interplay of diabetes, neuropathy, and ischemia increases the patient's susceptibility to infection. Importantly, the progression of infection, foot deformity, trauma, or lesion formations, may go undetected as a result of diminished sensation. Normal healing is often impeded due to multiple underlying mechanisms. These include the poor proliferation of fibroblasts due to diabetes, which delays the healing of connective tissue [2, 8]. Additionally, in hyperglycemic patients, super-oxide is excessively produced leading to oxidative stress which inactivates nitric oxide, an important molecule that promotes proliferation and angiogenesis, thus hindering the normal healing process of tissue and the formation of new blood vessels [3].

Routine inspection of the DF by a multidisciplinary team of clinicians, adequate glycemic control, patient/family education, and custom therapeutic footwear all play a pivotal role in proactive management and prevention of ulceration [8, 9]. Integrating wearable technology with these routine practices can provide continuous monitoring and real-time data, facilitating more bespoke and responsive clinical interventions. A recent review article [9], describes several instruments with sufficient validity and reliability for the assessment of diabetic neuropathy. These include sudomotor analysis, PAD detection via non-invasive electronic devices, wound 3D measurement; hyperspectral imaging for ulcer

prediction, and the probe-to-bone test for osteomyelitis diagnosis. Despite the availability of these assessment techniques, the frequency and cost of current screening methods pose notable obstacles. Several existing approaches are resource-intensive and may lack reliability, leading to restricted access and implementation. It is imperative to address these limitations to ensure timely and effective management of diabetic foot complications.

Due to its devastating health and socioeconomic impact, the literature is rich with articles related to all aspects of the DF, including prevention, assessment, and management. Mehmood et al. [10] reviewed wound dressings and automated monitoring technology designs, while Rajala and Lekkala [11] explored current available technologies for the assessment of plantar and shear stress. Lazarou et al. [12] examined the role of smart sensor technologies in the prevention and management of DFUs, focusing on their effectiveness in enhancing ulcer healing rates, reducing recurrence, and improving adherence to treatment protocols. A more recent article by Igiri et al. [13], also evaluated various therapeutic footwear designs, with particular emphasis on plantar pressure assessment. Similarly, Castro-Martins et al. [14] reviewed in-shoe plantar pressure measurement technologies for diabetic foot care, examining both commercial devices and emerging systems. Martín-Vaquero et al. [15] focused on previous work that addressed the correlation between temperature changes and ulcer development. Wang et al. [16], provided a detailed review exploring different plantar pressure and stress-sensing modalities, and presented commercially available sensory systems. Yuan et al. and Sun et al. [17, 18] reviewed flexible wearable sensors based on both optical and electronic sensing approaches. Okoduwa et al. [19] assessed the design and efficacy of therapeutic smart footwear for managing neuropathic DFUs.

Remote management and telerehabilitation of the DF have also been investigated by several research teams. Golledge et al. [20] reviewed the integration of a wider spectrum of risk factors with wearable and telehealth technologies to remotely manage the DF. Najafi and Mishra [21] summarized the latest trends related to the DF, focusing on the potential of remote management. Tekale et al. [22] examined novel interventions related to DF prevention, including assessment of the effectiveness of offloading footwear which monitors pressure and temperature, as well as exploring remote management and telehealth.

The current review aims to bridge the literature gap by exploring the integration of current state-of-the-art sensing modalities and smart wearables with novel actuation technology for real-time management and effective wound healing of the DF. Importantly, we focus on combining quantifiable mechanical and physiological/

wound-associated parameters which can predict/alert to the development of ulcers, healing scenarios, and means of tracking changes. We also explore telemedicine remote applications, as well as the role of 3D printing and smart materials in the design of insoles for wearable technology. Finally, both active and passive solutions used in the management of the diabetic foot are discussed and summarized. The remainder of this paper is organized as follows: Sect. “**Methodology**” outlines the employed search strategy, inclusion criteria, study selection process, and data extraction methodology. Sect. “**Sensorized wearables**” summarizes existing research on active and proactive approaches for diabetic foot prevention. It details current technologies for monitoring physical parameters (pressure, shear, temperature, and humidity) and biochemical parameters (pH levels and microcirculation) using smart wearable sensor systems. Additionally, it discusses other relevant attributes related to diabetic foot monitoring and management and includes an overview of vibrotactile actuation systems as emerging therapeutic interventions for symptom alleviation. “**Unsensorized wearables**” focuses on passive management solutions, highlighting innovations in smart materials, the integration of 3D printing technologies into wearable devices, as well as AI applications relevant to DF prevention and management. Lastly, Sect. “**AI technology and the diabetic foot**” presents concluding remarks, summarizing key findings, discussing limitations, and proposing potential developments in the field.

Methodology

Search strategy

The monitoring of parameters associated with the development of the DF and the use of dedicated sensing systems have recently become more critical with the expansion of the aging population and the exponential increase of T2D cases worldwide. With current healthcare resources already heavily burdened, advancements in wearable technology address significant gaps in current DF management by enabling continuous, non-invasive monitoring and providing real-time data to healthcare providers. Effective DF management is now accepted as an essential step for preventing ulcerations. Hence, a substantial body of literature has delved into the various parameters associated with inflammation and infection. We conducted a holistic survey to explore the state-of-the-art in terms of critical DF parameters, sensor modalities, wearables, and other tools and technologies currently used for diabetic foot monitoring and management. This review employed the academic databases Scopus and PubMed, focusing on the following search terms: diabetes, foot, sensor, management, monitor, pressure, shear, temperature, humidity, moisture, oxygenation, and pH level (parameter indicating acidity/alkalinity)

employing the following query algorithm ((Diabet* & Foot & Sensor*)|| (Diabet* & Foot & Manage*)|| (Diabet* & Foot & Monitor)|| (Diabet* & Foot & Pressure & Planar)|| (Diabet* & Foot & Shear)|| (Diabet* & Foot & Temperature)|| (Diabet* & Foot & (Humidity|| Moisture))|| (Diabet* & Foot & (oxygen*|| microcirculation))|| (Diabet* & Foot & pH)). The search was limited to scholarly articles published between 2012 and May 2023.

Inclusion criteria

The included articles were restricted to the following requirements: (1) written in English; (2) pertain directly to research as scholarly article document type; (3) published between 2012 and July 2024. 2,345 articles were identified to have met the defined criteria. To further strengthen the reliability of the review, additional selection included reputable, peer-reviewed journals, based on assessments of author credibility, methodological soundness, and consistency of reported findings. These measures were implemented to minimize the inclusion of unreliable literature and to ensure that only high-quality studies contributed to the review. Although a comprehensive formal quality assessment of all 98 included articles was not feasible, the rigorous multi-step screening process served as an effective safeguard for maintaining the scientific credibility and overall integrity of the review's findings.

Study selection and data extraction

To refine the search process and ensure the inclusion of the most relevant research body, an additional screening was applied by assessing the objectives and methodology of each included study to ensure alignment with the scope of this review. Duplicate papers were eliminated, and the remaining selected articles were those that either pioneered the creation of wearable sensor systems towards measuring specific relevant parameters or those that explored the use of smart materials and 3D printing in the design and development of wearables. Upon further screening, the final number of reviewed articles was reduced to 98 studies which collectively highlight the vast health challenges imposed by the diabetic foot and offer innovative solutions towards prevention, detection, and management. Figure 1(a) depicts the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flowchart, elucidating the following search and refinement process.

Consequently, a thorough analysis was conducted to explore related trends. Notably, the substantial increase in recent research, which particularly focused on the advancement of wearables for diabetic foot management was highlighted as shown in Fig. 1(b).

Moreover, in acknowledgment of the global attention to this critical topic, we incorporated multidisciplinary

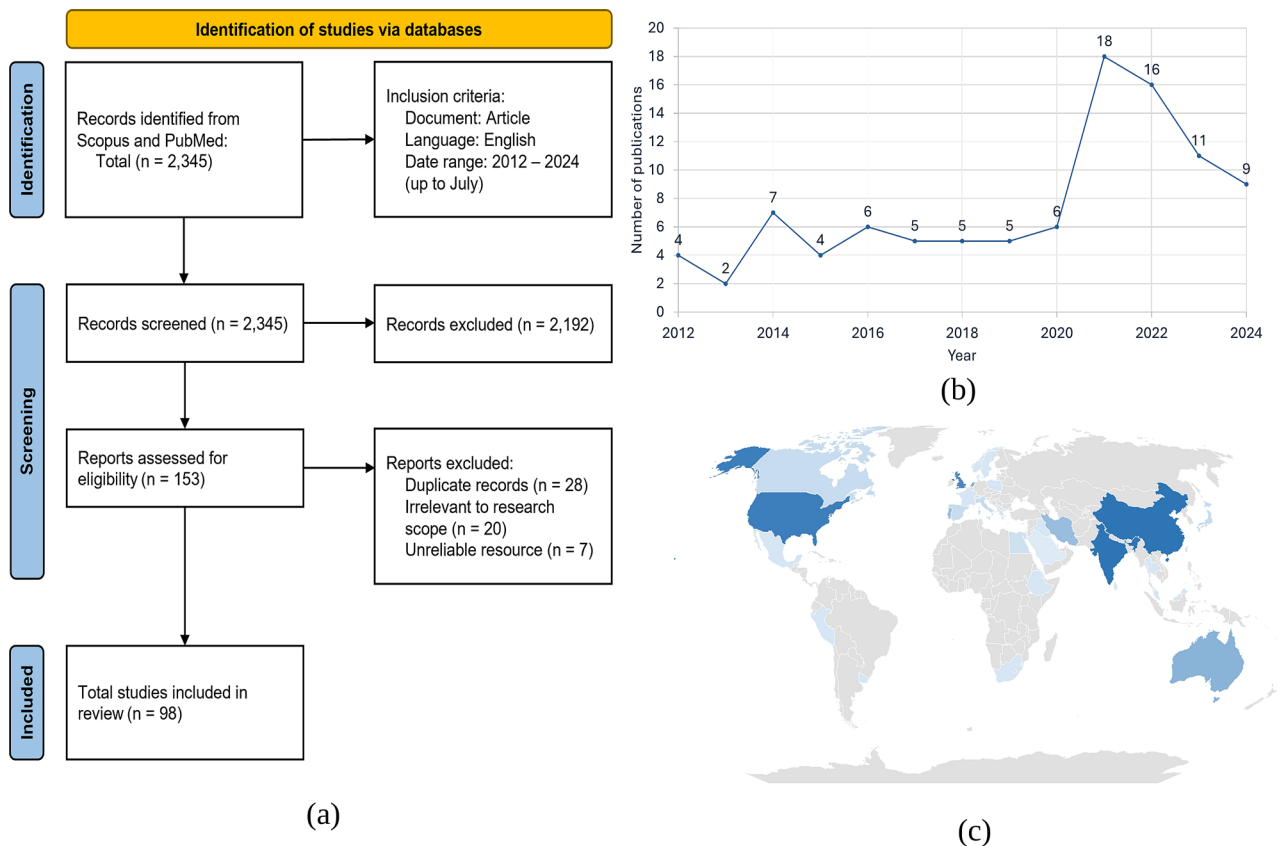


Fig. 1 (a) Search and screening process flowchart, (b) Publications per year, showing a notable increase in academic publications in recent years, particularly 2021 and 2022, (c) Publications worldwide with the highest concentration of publications in the United States, India, United Kingdom, and China

synergistic endeavours to collectively mitigate complications associated with the diabetic foot as illustrated in Fig. 1(c).

Sensorized wearables

The diabetic foot is a function of many complications, which are typically further amplified with the progression of associated diabetic nephropathy and peripheral vascular disease. This triad compromises the health of the lower limbs, often leading to limb loss [8]. It is therefore critical for clinical teams to continuously observe changes in the lower limbs/feet to allow for quick intervention and prevent the progression of complications. Accordingly, the novel preventative and management strategies discussed in this work focus on the important parameters critical for the identification of inflammation, infection, and poor wound healing. These include wound-related parameters, such as variations in the skin temperature and moisture level, reduced blood flow to the extremities, lack of hair growth, as well as any signs of cuts, blisters, or calluses [8]. Physical and chemical parameters include plantar pressure, magnitude and distribution, shear forces, temperature, humidity, microcirculation, and pH levels. Clinically, monitoring these parameters

provides critical insights into foot health and skin barrier function, enabling early detection of complications and facilitating timely and targeted interventions to prevent ulcer formations and promote wound healing. The holistic integration of these multifaceted parameters into foot wearables technologies can provide means for prevention and mitigation by continuously monitoring the dynamic variations, particularly in high-risk regions as depicted in Fig. 2(a). The regions of interest (ROI) identified in most studies focused on the foot's hallux, metatarsal heads, midfoot, and heel as shown in Fig. 2(b). These areas are typically subjected to high pressure and frequent shear forces which produce normal and shear stresses on the underlying tissue, hence rendering it susceptible to ulcerations. Moreover, it is important to note that the hallux, metatarsal heads, and heel are bony prominences further accentuating the risk of ulceration [23, 24].

Physical parameters and measures

Several physical parameters are associated with the onset and progression of foot ulceration, including plantar pressure magnitude and distribution, shear forces, temperature, and humidity. These attributes collectively provide important insight into the underlying foot condition,

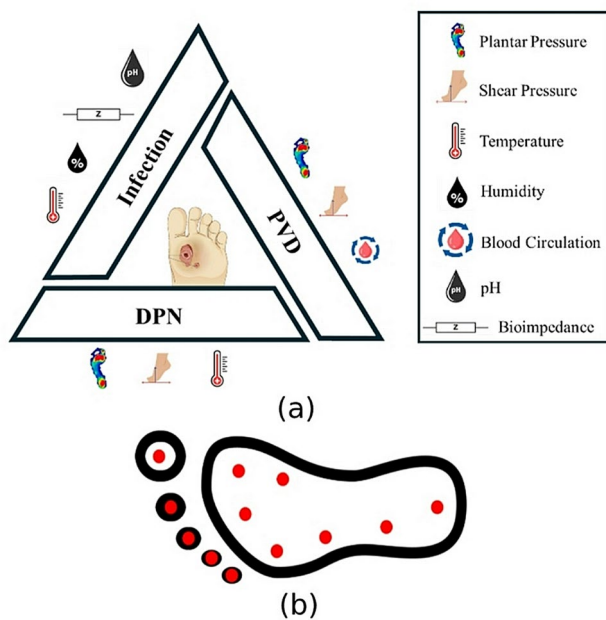


Fig. 2 (a) The triad of Infection, Peripheral Vascular Disease (PVD), and Diabetic Peripheral Neuropathy (DPN) and their related attributes, (b) ROI are more susceptible to ulcerations. They include the hallux, metatarsal heads, midfoot, and heel

and variations in their respective quantitative values can reveal potential foot complications, including inflammation and infection of subcutaneous tissue.

Plantar pressure and shear stress

Traditionally, plantar pressure has been considered among the key critical risk factors correlated to foot health and ulcer formation. High repetitive normal strain on the foot can result in injuries to the underlying tissue, leading to the development of ulcerations [23]. Consistent application of excessive pressure aggravates the condition of the injury and can greatly affect the gait of patients [24]. Additionally, the decreased sensation caused by peripheral neuropathy limits the patient's perception of the external environment and ulcer development.

Another equally important, but less explored factor due to assessment challenges is shear stress. The ground reaction forces (GRF) repetitively exerted on the lower limbs during weight-bearing activities, such as walking, result in cyclic loading in both normal and shear directions. These repetitive stresses have been shown to cause microtrauma, thereby accelerating the breakdown of tissue [25]. Accordingly, extensive research efforts have been devoted to the development of smart wearables and sensor systems for monitoring plantar and more recently shear pressure dynamics.

These modalities include capacitive [26–35], inductive [36–38], resistive [39–48], piezoresistive [49–52], piezoelectric [53], and more recently fiber optic-based sensing

mechanisms [54–61]. Each type of sensing technology presents advantages and drawbacks as elaborated next. In capacitive sensing systems, pressure is quantified by changes in the gap separating two conductive plates. This gap contains a dielectric material which changes in thickness when the pressure is applied resulting in capacitance variation.

In the system developed by Manikandan et al. [27], capacitive sensors were integrated into an insole to measure the foot's plantar pressure. A pressure distribution color mapping approach was employed to indicate areas of elevated pressure. In a more recent study by Luna-Perejón et al. [28], capacitive sensors were developed using a polydimethylsiloxane (PDMS) composite which served as a dielectric separating the capacitive plates as shown in Fig. 3(a). The PDMS layer also acted as a shock absorber due to its elastic deformation characteristics. The sensors were then integrated into a shoe to collect plantar pressure data in 12 locations for each foot, providing a suitable assessment modality. The main limitation reported in this study was the electric field generated by the body inducing disturbance in the collected values. On the other hand, these authors recommended the application of multiplication operations to partially remove the disturbance while retaining important gait information. Tang et al. [26] also employed a capacitive sensing mechanism to investigate both normal pressure and shear stress as critical factors associated with ulcer formation. This work offered an innovative sensorized insole capable of normal pressure and shear assessment.

Several other studies employed capacitive sensing methods to analyze pressure distribution patterns through commercially available systems, such as Pedar and Pliance [62, 63]. Raspovic et al. [29] used the Pedar system to validate an affordable, easy-to-use DF pressure relief shoe by evaluating its efficacy in redistributing the pressure away from areas susceptible to ulceration. Similarly, in the study conducted by Begg et al. [30, 31], both plantar and shear pressure were quantified and monitored for patients who had already developed ulcers. In these cases, immediate care is critical to reduce the pressure on the ulcerated sites using a total contact cast (TCC), an effective intervention method commonly employed for pressure reduction. Using capacitive measuring systems (Pedar and Pliance), Begg et al. monitored the pressure between the foot and shoe and measured the contact area, pressure, as well as the maximum forces between the TCC and the anterodistal and posterolateral-distal regions subjected to the highest loads, respectively. Withers et al. [32] also employed a Pedar system to assess the offloading effects of a removable cast walker (RCW). In this study, the authors investigated the impact of plantar pressure, reporting on the efficacy of pressure reduction for diabetes-related foot

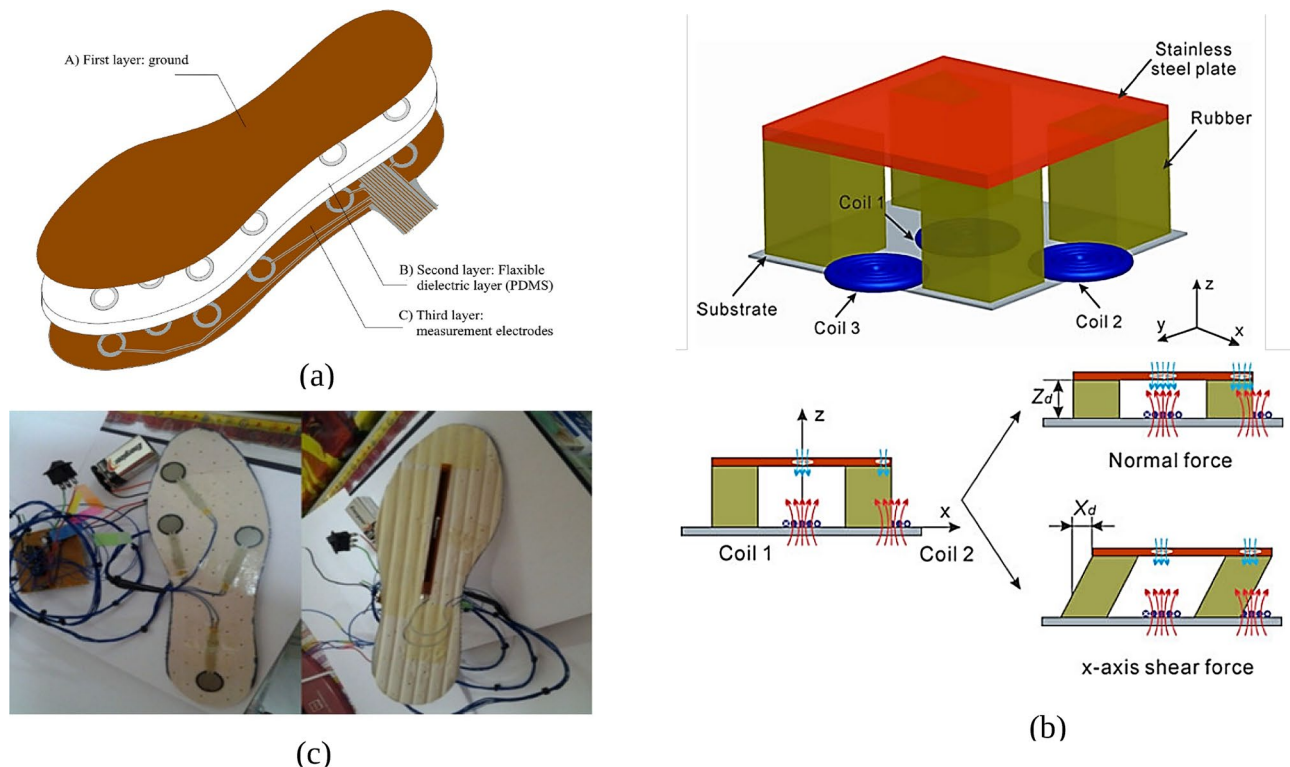


Fig. 3 Various sensing mechanisms used in wearables for the management of diabetic foot (a) Scheme of main layers that make up the proposed plantar insole capacitive PDMS [28] CC-BY license, (b) Pressure and shear monitoring using three-coil induction force sensor [36] ©2015 IEEE, (c) Prototype system proposed to monitor and analyze gait by looking specifically at the foot force and the plantar flexion using FSR and FBS sensors [39] CC-BY license

ulcerations. Van Netten et al. [33] adapted the capacitive Pedar system to investigate plantar pressure for indoor vs. outdoor customized footwear while also studying the adherence of high-risk patients to the prescribed shoes. This study found that both indoor and outdoor footwear provided similar offloading effects. On the other hand, indoor custom-made footwear resulted in significantly higher patient adherence. In another work by Zwaferink et al. [34], the Pedar system was employed for the development of customizable footwear aimed at managing high plantar pressure. High-risk areas were evaluated to recommend important modifications which can be applied to effectively reduce the peak pressure and optimize the footwear. Notably, multiple studies have shown the significance of manufacturing patient-specific footwear customized to the shape, size, and plantar pressure distribution of the patient's foot to reduce elevated pressure. Forogh et al. [35] conducted a comprehensive study to test the reliability of the Pedar system between different days. While the authors confirmed its accuracy and reliability, they demonstrated that with increased neuropathy, the system's reliability tends to decrease in between-day tests.

Inductive sensing relies on electromagnetic fields for the detection of surrounding conductive objects. The movement of these objects causes variations in the

electromagnetic induction proportional to the applied changes in forces, allowing the measurement of plantar pressure [36]. Du et al. [36] developed a low-cost inductive sensor to enable the simultaneous measurement of normal forces applied on the plantar surface, as well as the two-axis shear forces as shown in Fig. 3(b). The sensor's small size rendered it practical, as it could be easily embedded into footwear for continuous real-time monitoring. In another study conducted by Wang et al. [37], a shear load inductive plantar sensing (SLIPS) system was developed and integrated into an insole to measure both plantar and shear pressures across the foot. The results of this study verified that the SLIPS system operated efficiently under dynamic gait loading. Further studies also reported the design of tri-axis force sensors which concurrently quantify both normal and shear loads on the plantar surface. Wang et al. [38] investigated the performance of such sensor for different configurations using a computational finite element model, concluding that using four-square coils with the maximum number of turns yielded the optimal sensor configuration.

Resistive sensors are also commonly employed in measuring foot pressure. These leverage the principle where mechanical forces exerted on these sensors causes the conductive particles to move closer, hence resulting in a decrease in the electrical resistance proportional to the

applied force. The use of this measurement technique provides a low-cost dynamic approach for measuring the ground reaction forces (GRFs). In a study conducted by Anas [39], a resistive-based prototype system was proposed to monitor and analyze gait by assessing the GRFs as shown in Fig. 3(c). The system is comprised of a Force-Sensitive Resistor (FSR) and a Flexible Bend Sensor (FBS) to measure the foot forces and flexion, respectively. On the other hand, while these sensors are small in size and cost-effective, they tend to exhibit low accuracy and resolution [39]. It is also essential to highlight the importance of FSR sensor placement and its impact on accuracy, as well as the selection of suitable materials which can greatly affect the measured forces. FSR sensors were also used within feedback systems to ensure that the measured peak plantar pressure is maintained within a set threshold, thereby facilitating the customization of therapeutic footwear and enabling patient-specific adjustment [40]. Likewise, Bose et al. [41] reported the development of a smart insole incorporating 11 FSR sensors within silicone inserts designed to measure plantar pressure. The study demonstrated the insole's efficacy in offloading plantar pressure, highlighting its potential in preventing ulcer progression.

In a study by Ferber et al. [42], the commercially available pressure-sensing array, Orpyx [43], was integrated into an insole to provide real-time feedback. This is of notable significance for neuropathy patients who exhibit limited sensation in the lower limbs towards ulcer and limb loss prevention. Similarly, Matijevich et al. [44] employed Orpyx technology to monitor plantar pressure and temperature, while simultaneously tracking patient step count and adherence. Another study that utilized the Orpyx intelligent insole added an alerting system that notifies the patient of increased pressure. The results showed that alerting patients led to a 71% reduction in ulcers. Additionally, this research team found that continuous feedback reduced the plantar pressure by inducing a neuro-learning response where users were trained to maintain posture that led to pressure variability on the plantar surface of the foot [45, 46]. Further studies employed resistive sensor-based systems, such as the Fscan (Tekscan Inc., USA) [64] to study the impact of footwear on plantar and shear loading during gait at various walking speed [47, 48].

In contrast to the previously discussed sensing methods, piezoresistive sensors are devices fabricated from materials that exhibit changes in electrical resistance in response to applied mechanical stress due to the piezoresistive effect. This technology was adopted in the development of the WalkinSense, a portable plantar pressure measuring system proposed by Healy et al. [49]. Perrier et al. [50] also used piezoresistive sensing, where a smart diabetic sock was custom-designed to accurately measure

plantar pressure in real-life conditions. This customizable, wireless, yet washable prototype has the additional capability to send notifications to the user/clinician to raise alerts in case of ulceration risk. The integration of a warning system in wearables can significantly mitigate the drastic progression of ulcer formation by prompting immediate action to control and minimize risks. Tiwari et al. [51] developed a cost-effective insole composed of a polyester/nylon blend fabric, designed to measure plantar pressure in individuals with diabetes. The insoles encompassed 31 piezoresistive sensors, each capable of detecting pressures up to 960 kPa. The design emphasizes flexibility and durability, ensuring both user comfort and accurate pressure sensing during daily activities. Similarly, Wang et al. [52] developed a novel low-cost flexible sensing system that provided continuous long-term monitoring on a daily basis. The insole included a high-sensitivity pressure sensor array that tracked and transmitted the data wirelessly, hence displaying the changes in real-time through a mobile application.

Similar to piezoresistive sensors, piezoelectric sensors also leverage the properties of specific materials to generate electricity in response to mechanical stress. On the other hand, they produce electrical charges instead of resistive variations when deformed. Klimiec et al. [53] reported on the development of a plantar pressure diagnostic system to analyze the distribution of foot pressure during. According to the authors, this design offered many advantages, including reliability, durability, and user-friendliness.

Optical Fiber Sensors (OFS) present a unique sensing mechanism capable of quantifying different parameters with associated DF pathology. Fiber Bragg Grating sensors (FBGs), acting as a wavelength selective filter, consist of an optical fiber segment with periodic variations of the refractive index fiber core. These variations form the grating structure, which reflects specific wavelengths of light (Bragg wavelength). When the fiber is exposed to light, wavelengths that do not abide by the Bragg condition are transmitted, while those that meet the Bragg condition are reflected. Therefore, when an external stimulus, such as strain/ pressure or temperature, is induced in the structure containing the Bragg sensor, the Bragg wavelength shifts triggering a change in the reflected wavelength peak. This shift can be monitored and analyzed to quantify the interrelated stimulus [54, 55]. Figure 4(a) demonstrates the working mechanism of FBGs.

Numerous studies have explored the use of FBGs in monitoring plantar pressure, as demonstrated by Suresh et al. [54]. In this work, a shoe was equipped with two sensors attached at the forefoot and hindfoot to monitor the applied pressure while walking at different speeds. The results indicated that at a higher speed, the rate of pressure increases in the forefoot as compared to the

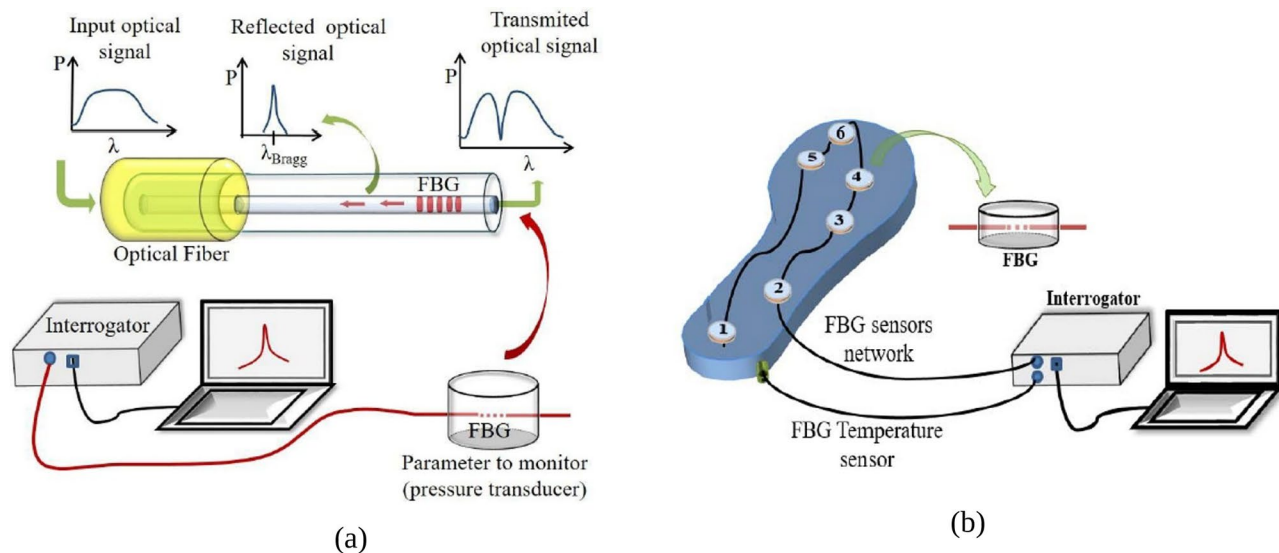


Fig. 4 (a) The sensing mechanism of FBG sensors [55] ©2019 IEEE, (b) A diagram showing FBG sensors integrated into a shoe insole [55] ©2019 IEEE

heel. Further investigations comparatively evaluated the use of FBGs and lead zirconate titanate (PZT) piezoceramic sensors, emphasizing the advantages of FBGs as a sensing modality [56]. These authors also developed a system consisting of an FBG sensor integrated into a pad to monitor static plantar pressure, a solution that could be used as a stand-alone independent platform or integrated into a standard shoe [57]. In another work by Domingues et al. [58], an insole was designed and validated using a network of multiplexed FBG sensors. The system provided accurate and consistent data when measuring the induced vertical ground reaction forces during gait. The correct placement of sensors on key points on the foot was highlighted as key to infer vital information regarding the foot's structure and health condition as shown in Fig. 4(b). Subsequent studies by the research team involved the integration of two FBG sensors in the same local sensing structure, and along the same optical fiber to simultaneously measure the shear and vertical forces during gait [55, 59]. Further enhancements were incorporated into the design, such as utilizing 3D printing to produce the insole base and the body of the sensor cell [60]. This work was further refined by providing full gait and means of real-time monitoring via an edge-cloud networking architecture [61]. Compared to their electronic counterparts, FBG sensors offer a superior sensing modality due to their compactness, robustness, durability, minimal signal processing requirements, negligible weight, linearity, and electromagnetic insensitivity [56, 58].

Other alternative sensing systems used to measure the critical parameters associated with DF include a study that used a microstrip patch antenna sensor [65]. This sensor provided shear pressure assessment, where foam

substrate deformed in response to shear loads, such that changes in the length were detected by a shift in the frequency of the antenna. The authors suggested that the high sensitivity, compact size, and fine resolution of this sensor make it suitable for integration into a footwear sensory system for DF ulceration monitoring.

Ren and Liu [66] proposed a low-cost insole system based on photoelectric sensing devices to measure plantar pressure distribution. Using this technology, the authors were able to gather important data about gait and posture through analysis of the insole's deformation. Heck et al. [67] also reported on the development of a sensing system which monitors pressure using a sensor integrated into a bandage. The systems included an alert system, where elevated pressure values exceeding a preset threshold triggered an alert that directly notified the user.

In a study conducted by Du et al. [68], the commercially available LEGSys and BalanSens wearable technologies [69] were used to closely examine gait and balance parameters in an elderly diabetic population. Similarly, Wrobel et al. [70] assessed gait and balance using the LEGSys, which was integrated into a shear-reducing insole. Table 1 summarizes published work in plantar pressure and shear monitoring.

Temperature

Another important parameter linked to the onset of diabetic foot complications is temperature. Multiple studies have explored the significance of temperature in predicting the development and progression of pressure ulcers or other chronic problems of the foot. Research findings indicate that a 1.2 °C increase in temperature within 24–96 h typically precedes the development of an ulcer

Table 1 Published work in plantar pressure and shear monitoring

Ref	Sensor Type	Wearable	Measured Parameters	Num. of sensors per foot	Clinically tested		Results
					Patients	Healthy	
[26]	Capacitive	Insole	Plantar pressure and shear	4 sensors	✓	✓	The designed system has sufficient sensitivity and can effectively help assess risk of ulceration.
[27]	Capacitive	Insole	Pressure distribution		✓		Changes in the capacitance due to variations in the pressure impact the output voltage which can be visualized through color mapping.
[28]	Capacitive	Insole	Pressure	12 sensors			The acquired sensitivity up to 0.76 fF/Kpa.
[29]	Capacitive (Pedar-X)	DH Pressure Relief Shoe containing the Pedar in-shoe		99 sensors	✓		The DH Pressure Relief Shoe effectively reduced the plantar pressure by 51% in comparison to control shoe and by 43% in comparison the user's regular footwear.
[30]	Capacitive (Pedar-X & Pliance)	Total contact cast (TCC)	Plantar load and TCC load	99 & sensor strips in different locations		✓	The average load exerted on the TCC wall per step was equivalent to 23 – 34% of the average plantar load on the opposite foot.
[31]	Capacitive (Pedar-X & Pliance)	Total contact cast (TCC)	Plantar load and TCC load	Pedar-X: 99 pressure sensors Pliance: 2 sensor strips	✓		The TCC bears a significant amount of the load, as indicated by the reduced plantar contact area compared to the shoe-cast (reduce and redistribute pressure).
[32]	Capacitive (Pedar-X)	Insole	Plantar pressure	99 pressure sensors	✓		Peak plantar pressures were lower in cases where the RCWs were used.
[33]	Capacitive (Pedar-X)	Custom made footwear	Plantar pressure	99 pressure sensors	✓	✓	Increased indoor adherence (48–77% after 12 months of usage)
[34]	Capacitive (Pedar-X)	Custom made footwear and insoles	Plantar pressure	99 pressure sensors	✓		Personalized footwear significantly decreased the MTH peak pressure.
[35]	Capacitive (Pedar-X)	Pedar in-shoe system	Plantar pressure	99 pressure sensors	✓		Confirmed reliability of Pedar system.
[36]	Inductive	Force sensors attached at forefoot and heel	Normal force and shear forces (in x and y direction)	3 coil sensors		✓	High sensitivity of developed prototype.
[37]	Inductive (SLIPS)	Insole	Plantar and shear loading	64 sensing nodes			Sensitivity of SLIPS system $F_z = \pm 0.08$ N, $F_x/F_y = \pm 0.02$ N
[38]	Inductive	Force sensor	Plantar normal and shear load				The prototype achieved high performance with a shear load detection of ± 16 N and normal load of 105 N.
[39]	Resistive (FSR & FBS)	Insole	Ground reaction force based on foot force and plantar flexion	4 FSR sensors and 1 FBS sensor		✓	The developed prototype can monitor clinical gait.
[40]	Resistive (FSR)	Insole	Peak plantar pressures	7 FSR sensors	✓		A pressure monitoring system was developed with a biofeedback feature.
[41]	Resistive (FSR)	Insole	Plantar pressure	11 FSR sensors		✓	Developed insole effectively offloaded plantar pressure.
[42]	Resistive (PSRU by Orpyx vs. Pedar-X)	Insoles	Plantar pressure	Array of 8 sensory units vs. 99 sensors		✓	The developed PSRU yielded similar data to that of GS-PSD.
[44]	Resistive (Orpyx)	Insole system	Plantar pressure and temperature	Array of 22–37 FSR sensors for plantar pressure and 5 temperature sensors	✓		The custom sensory insole tracks the plantar pressure, and temperature along with the step count and the patient adherence.

Table 1 (continued)

Ref	Sensor Type	Wearable	Measured Parameters	Num. of sensors per foot	Clinically tested		Results
					Patients	Healthy	
[45]	Resistive (Orpyx)	Insole system	Plantar pressure	8 pressure sensors	✓		A significant reduction of ulcer incidence was noted for patients who abided with the use of an offloading-insole system.
[46]	Resistive (Orpyx)	Insole system	Plantar pressure	8 sensors	✓		Higher plantar pressure was seen in the control group across all areas of the foot.
[47]	Resistive (F-scan & shear force sensor)	Shoe	In-shoe plantar pressure and shear	10 F-scan sensors and a shear force sensor	✓		The developed system has the ability to reduce the pressure and shear force.
[48]	Resistive (F-scan & shear force sensor)	Insole	Plantar pressure and shear force	10 F-scan sensors and a shear force sensor	✓	✓	The shear sensor provided a 0.1 Kgf accuracy. The study suggests the importance of reducing applied stress on the metatarsal bone region.
[49]	Piezoresistive (Walkin-Sense) vs. F-scan	Insole	Plantar pressure	8 Walkinsense sensors and F-scan sensors		✓	Confirmed the effectiveness of WalkinSense system in measuring and reporting plantar pressure.
[50]	Piezoresistive	Socks	Plantar pressure	8 sensors		✓	System measures the applied pressure and transmits the acquired data to a smart phone.
[51]	Piezoresistive	Insole	Plantar pressure	31 sensors			The sensor array covers an active area of 78.5 mm ² and has a sensing range of up to 960 kPa.
[52]	Piezoresistive	Shoe system	Plantar pressure	8 sensors	✓	✓	Plantar pressure data acquired from the developed system was used to classify healthy, diabetics without neuropathy, and diabetics with neuropathy.
[53]	Piezoelectric	Insole	Plantar pressure	8 sensors			System can be used for diabetic foot examination and monitoring recovery post injury.
[54]	Optical	Shoe system	Plantar pressure	2 arc-shaped FBG sensors		✓	Arc shaped technique yielded an average pressure sensitivity of 1.3 pm/kPa and a resolution of 0.8 kPa.
[55]	Optical	Insole	Plantar pressure	6 FBG sensors			An optical fiber-based system was developed with multiplexed FBGs to monitor the plantar pressure. The system elements (FBG2 and FBG3) yielded a sensitivity coefficient of 7.6 ± 0.23 pm/kPa and 8.3 ± 0.24 pm/kPa.
[56]	Optical	Shoe sole	Plantar pressure	FBG and PZT		✓	FBG pressure sensors sensitivity and resolution of 1.3 pm/kPa, 0.8 kPa, respectively. PZT sensors sensitivity and resolution of 7.06 mV/kPa, 0.14 kPa, respectively.
[57]	Optical	Insole	Plantar pressure	4 arc-shaped FBG sensors		✓	A pressure sensitivity configuration averaging 1.2 pm/kPa and a resolution of 0.8 kPa.
[58]	Optical	Insole	Vertical ground reaction forces	6 FBG sensors		✓	The devised platform enables monitoring of applied plantar forces with a sensitivity of up to 11.06 pm/N.
[59]	Optical	Insole	Plantar pressure and shear forces	5 points each containing 2 FBGs		✓	The system presented the following sensitivities for vertical (Kv) and shear (Ks) forces: K1V = $(14.15 \pm 0.10) \times 10^{-3}$ nm/N K2V = $(7.35 \pm 0.02) \times 10^{-3}$ nm/N K1S = $(-26.02 \pm 0.08) \times 10^{-3}$ nm/N K2S = $(-24.29 \pm 0.08) \times 10^{-3}$ nm/N
[61]	Optical	Insole	Gait analysis including foot plantar pressure	8 FBG sensors		✓	A full gait analysis optical fiber-based system was presented. Table 1 presents the sensitivities to vertical force, angle and strain, and temperature.
[65]	Shear deformation effects on resonant frequency	In-shoe shear sensor	Plantar shear	Antenna shear sensor			The antenna sensor exhibited a shear deformation sensitivity estimated at 5.8 MHz mm ⁻¹ . The sensor can detect shear displacements as minute as 17 μm.
[66]	Photoelectric sensing	Insole	Plantar pressure	Pressure sensor unit		✓	The photoelectric-based pressure sensing insole effectively detects the pressure distribution by monitoring the insole deformation during movement providing information about the gait.

Table 1 (continued)

Ref	Sensor Type	Wearable	Measured Parameters	Num. of sensors per foot	Clinically tested		Results
					Patients	Healthy	
[67]	Pressure sensitive film	Foam bandage	Plantar pressure	Pressure sensor placed accordingly (bandage)		✓	A wireless pressure sensor bandage was introduced providing continuous monitoring and alerts when the set pressure threshold is exceeded.
[68]	LEGSys & BalanSens	Sensor unit	Gait and balance parameters	2 Sensors attached to the lower back and shin.	✓		The platform demonstrates effective monitoring in assessing the gait and balance changes
[70]	LEGSys	Insole	Gait, balance, and temperature		✓		The developed shear-reducing insole can significantly minimize the temperature changes.

due to changes in metabolic activity related to inflammation or infection [71]. Moreover, a 2.2 °C rise in temperature of an area, relative to the same contralateral region on the other foot, can indicate an underlying inflammation of the tissue, while a decrease in temperature can suggest ischemia [72]. Accordingly, vigilant monitoring of skin temperature has the potential to greatly enhance the prediction and mitigation of diabetic foot ulcers and other DF complications. While conventional measurement techniques can roughly quantify the temperature of the skin, they present several drawbacks, including limited accuracy and reliability, and poor adherence. Alternatively, various studies explored flexible, non-invasive, yet accurate temperature sensors which could provide real-time monitoring of the foot's environment as outlined in Table 2 at the end of this subsection [71, 73].

In a study conducted by Reyzelman et al. [74], a reusable sock with an embedded sensor was developed for daily temperature assessment. The wireless system continuously

monitored changes and promptly notified the patient regarding temperature increase towards early detection of ulceration and Charcot's foot, hence allowing for timely intervention. This study further investigated the correlation of skin temperature variations towards the early detection of DF ulcerations prior to noticeable wound development. Embedding the sensor into the base of a sock, outlier temperature variations were identified, and the developed system was able to predict ulceration 15 days prior to observable injury [75]. Issac et al. [76], also proposed a system that accurately quantifies the foot temperature and integrates the sensor seamlessly into the shoe design. Similarly, Schmalen and Meyer [77] developed two smart insoles equipped with temperature sensors to diagnose diabetic foot ulcers. In their study, a negative temperature coefficient (NTC) sensor and a fully printed resistance temperature detector (RTD) were tested demonstrating an accuracy of 0.5 °C. The study concluded that both sensors were effective and recommended future exploration of additional sensors and

refining the calibration system to enhance diagnostic capabilities.

From a different perspective, Brooks et al. [78] examined the cost-effectiveness of remote foot temperature monitoring using the commercially available Siren Care diabetic sock. This system provides reliable continuous tracking of foot temperature through an application that notifies the user and sends real-time information directly to the clinician.

Importantly, monitoring temperature has another vital role in DF management in terms of overseeing the adherence of patients to prescribed footwear [79]. Lutjeboer et al. reported patient adherence to orthopaedic footwear by incorporating an orthometer into a shoe to assess the temperature of the foot. Keukenkamp et al. [80] also investigated short and long-term adherence to custom-made footwear tailored to effectively prevent foot ulcerations by redistributing the pressure to lower the mechanical stress applied to high-risk regions. In this study, a temperature sensor was placed in the footwear to record the temperature in 1-minute intervals to quantify the user's adherence. Improved compliance was associated with indoor custom-made footwear.

In another study by Jarl and Tranberg [81], pressure and temperature sensors were integrated into a therapeutic shoe to observe whether sealing a shoe can positively offload the pressure and help heal diabetic foot ulcers. The pressure was monitored using the Fscan system while the temperature was tracked to assess users' compliance. This study confirmed the feasibility of the system in offloading and expediting the healing of forefoot ulcers.

Other researchers, including Reddy et al. [82] explored assessing temperature variations during walking. This system used four sensors to measure the temperature while simultaneously monitoring the pressure using an Fscan, indicating an increase in foot temperature during walking due to the increase of metabolic demand. Notably, a 1 °C increase in temperature has been associated with a 10% increase in tissue metabolism [82]. Metabolic

Table 2 Wearable technology that monitors temperature and pressure

Ref	Wearable	Measured Parameters	Num. of sensors per foot	Clinically tested		Results
				Patients	Healthy	
[72]	Sock	Temperature	8 sensors	✓		A smart sock was developed to track temperature changes at the ROI.
[74]	Sock	Temperature	6 sensors	✓		The sock provided readings consistent with the reference standard. With variations of no more than 0.2 °C, demonstrating reliability of system.
[75]	Sock (Siren)	Temperature	6 points	✓		System detected a change (up to 4 °F) to the contralateral foot 15 days preceding an injury.
[76]	Shoe system	Temperature			✓	The sensor is developed with a 0.01 °C accuracy.
[77]	2 Insoles	Temperature	7 sensors			Both types of insoles can achieve a temperature accuracy below 0.5 °C.
[78]	Sock (Siren care)	Temperature	6 sensors			Remote foot temperature monitoring systems provide a cost-effective solution to patients as it can greatly improve their quality of life and reduce expenses.
[79]	Insole	Temperature	2 sensors		✓	The sensor underwent testing and validation, confirming its suitability as a system for temperature monitoring.
[80]	Custom-made footwear & Pedar-X	Temperature and pressure	1 temperature sensor 99 pressure sensors	✓		The customized footwear significantly increased indoor footwear adherence.
[81]	Custom-made footwear & F-scan	Temperature and pressure	1 temperature sensor F-scan pressure sensors	✓		Successful healing of ulcers and confirmed feasibility of the system in managing forefoot ulcers.
[82]	Insole (temperature) & F-scan	Temperature and pressure	4 temperature sensors F-scan pressure sensors		✓	Observed fluctuations in temperature during walking have the potential to augment tissue metabolism where a 1 °C rise results in a 10% increase in tissue metabolic rate.
[83]	TAPMARI (Temperature and Pressure Monitoring and Regulating Insole) & Pedar-X	Temperature and pressure		✓	✓	The presented system showcased temperature regulation for optimal conditions within the footwear. By maintaining a temperature around 28 °C (± 1 °C), the system helps reduce metabolic demands on the foot and prevents cell autolysis by restoring oxygen balance.
[85]	Insole	Temperature and pressure	1 thin-film NTC temperature sensor 1 FSR pressure sensor		✓	The developed device offers a cost-effective system for monitoring temperature and pressure.
[86]	Insole	Temperature and pressure	1 infrared (IR) temperature sensor 4 Flexi force sensors			The device prototype monitors the temperature and pressure with confirmed accuracies of 99.05% and 99.30%, respectively. It also integrates red light therapy for therapeutic benefits.
[87]	Sock	Temperature and pressure	7 NTC temperature sensors MatScan plantar-pressure measurement system		✓	The developed temperature measuring system was able to accurately predict pressure changes due to readings acquired from two sensors at the medial and lateral midfoot regions.

demand during walking requires adequate blood flow, the lack of which might cause ischemic damage.

Yavuz et al. [83] highlighted the significance of maintaining physiological tissue at cooler temperatures to prevent cell autolysis. Using a previously designed temperature and pressure monitoring and regulating insole (TAPMARI) system, they employed thermostat sensors to regulate the foot's temperature. Their results demonstrated the ability to effectively decrease ulceration rates

and avoid complications, such as amputation and limb loss.

Similarly, several other researchers designed and developed various insole systems for dynamic monitoring of temperature and pressure [84]. Several off-the-shelf pressure and temperature sensors were assessed and compared in terms of cost, response, and range. This comprehensive evaluation identified the FSR sensor and NTC thermistor as suitable for pressure and temperature assessment, respectively. Thorne et al. [85] reported on

the development of an insole which utilizes a flexible FSR pressure sensor and a negative temperature coefficient (NTC) thermistor to simultaneously monitor pressure and temperature. The readily available systems, Fscan and Flir thermographic camera, were used to correlate the gathered data. The authors suggested that this system offers a cost-effective alternative for monitoring pressure and temperature, as it can successfully aid in detecting early complication signs in high-risk cases.

In another investigation led by Kumsa et al. [86], a system was proposed to monitor pressure and temperature while providing therapeutic interventions through red light treatment. This distinctive feature was said to enhance cellular energy synthesis, hence prompting more effective restructuring of elastin fibers and fibroblast collagen. It also promotes the simulation of new fiber production which is needed for wound healing. Penetrative radiation has the potential to cure calluses while actively promoting the release of nitrous oxide, a relevant compound for vasodilation.

González et al. [72] developed a wearable sock to track foot temperature variations at specific regions of interest (ROIs). These included the first toe, first and fifth metatarsal heads, medial foot, and heel. This system utilized a thermistor, particularly an NTC sensor, to track the temperature of the foot during gait. The collected data, stored on a smartphone, provided an alert system to notify patients as necessary. However, these authors acknowledged the need for further investigation to test the footwear in longer walks, including outdoor experiments; improve the comfort of the sock; and importantly consider the influence of foot sweat on the collected data. Another smart sock system developed by El-Nahas et al. [87] employed seven thermal NTC sensors intricately embedded into the fabric of a sock to evaluate the plantar pressure and temperature during normal daily life activities.

Humidity

It is well known that the microclimate of the foot can be influenced by humidity, a critically important factor in the etiology of ulcer development. Elevated skin moisture can cause adverse effects on the epidermis (outer layer of the skin), particularly the stratum corneum, leading to skin breakdown [71]. Moreover, several studies have established the correlation between skin moisture and friction, where excessive skin dryness increases the coefficient of friction, potentially leading to skin damage and fissures [71]. From a wound-healing perspective, it is imperative to maintain optimal humidity levels to establish an appropriate wound-healing environment. An excessively wet environment can lead to maceration (softening and breaking down of skin resulting from prolonged exposure to moisture), while a dry microclimate

can disrupt the healing process by desiccating the wound (separation of a wound due to improper wound healing). Effective humidity management has been shown to speed up recovery and healing of open wounds [73].

Several research studies have investigated humidity as a key factor in the management of DF, alongside temperature and pressure, as summarized in Table 3. Yik et al. [88] investigated pressure, temperature, and humidity using five different types of socks. They identified the rearfoot (RF) as an area of exceptionally high pressure in comparison to other regions. This study demonstrated that friction was dramatically reduced when any type of sock was worn and highlighted the importance of maintaining a stable environment by controlling the temperature and humidity to protect the sole from bacterial and fungal growth. The drastic effect of the sock's fabric on the microclimate of the foot in terms of temperature and humidity was also noted.

In another study conducted by Milne et al. [89], WoundSense moisture commercial sensors [90] were used to observe the moisture condition of the wound during the change of dressing. These authors reported that only 44.9% of wounds retained optimum moisture during dressing changes, while 81% of the nurses interviewed reported that dressing changes could be very painful due to the adhesion of the dressing's matrix to the dry wound. Sandoval-Palomares et al. [91] developed an insole capable of assessing the microclimate of the foot by monitoring the temperature and humidity of the sole. Moulaei et al. [92] also developed a system to assess pressure, temperature, and humidity, integrated with an application for real-time monitoring. This system alerted diabetic patients, thereby facilitating self-management. Another study by Pascali et al. [93] presented an insole system that monitors temperature, pressure, and moisture, where the latter was specifically used to examine glucose levels in sweat. The system incorporated hardware and software components as well as finite element analysis (FEA) simulation, to evaluate temperature.

Biochemical parameters

Other factors that play a pivotal role in diabetic foot pathology include chemical parameters that govern the microcirculation, as well as the pH of the skin. Table 4 summarizes previous research examining the biochemical parameters including microcirculation and pH as well as other attributes.

Microcirculation

Within the realm of factors which could indicate diabetic foot progression is microcirculation. This parameter influences various aspects of the microvascular system including tissue perfusion, fluid dynamics, and oxygenation. In the context of the diabetic foot,

Table 3 Technologies that explore humidity

Ref	Wearable	Measured Parameters	Num. of sensors per foot	Clinically tested		Results
				Patients	Healthy	
[88]	5 different socks	Humidity, temperature, and pressure (Pedar-X)		✓	✓	The 5 tested socks effectively reduced the plantar pressure. However, variations in the foot skin temperature and humidity regulation were observed for different knitting structures and fabrication.
[89]	Wound-Sense sensor (used with dressings)	Moisture		✓		The integrated sensor measures the moisture at the wound site to indicate when the dressing requires change.
[91]	Sensor placed into an insole	Microclimate (temperature and humidity)	5 points–10 sensors		✓	Humidity: 100% ±2% Temperature: 40 °C to 123.8 °C ±0.3 °C
[92]	Insole	Humidity, temperature, and pressure (Pedar-X)	4 FSR pressure sensors 1 temperature and humidity Pedar-X	✓	✓	Humidity: 0–100% ±2% Temperature: -40 °C to 125 °C ±0.3 °C Pressure: 0.1–10.0 ² N
[93]	Insole	Glucose in sweat and pressure	8 points			Glucose in sweat: 0–6 mM (0–108 mg/dL) glucose concentration optimized linear response (for sweat-sensing), with a LOD of 3.27 M (0.058 mg/dL) and a sensitivity of 21 A/mM cm ² in the PBS solution Pressure: The figure of merit for the full custom pressure sensors is 4.7 ± 1 GPa.

observable endothelial damage and microvascular dysfunction lead to ulcerations. Understanding the microcirculatory function is very important as it sheds light on mechanisms of injury, inflammation, and repair [94]. Potentially, the assessment of microcirculation, in conjunction with other relevant factors previously discussed in this review, facilitates early detection of diabetic foot-related issues, hence enabling timely intervention.

Several studies have explored different DF associated microcirculation aspects. Yu et al. [95] particularly focused on edema, which refers to the fluid buildup in the lower extremities. In patients with DF, elevated glucose levels in the blood can cause arterial thickening, which leads to loss of tissue elasticity, impairing the necessary circulation to the lower limbs. In the event of an injury, a physiological defense mechanism is initiated, triggering an inflammatory response through the lymphatic system and blood vessels. Poor drainage can cause accumulation of fluids leading to pain and limiting mobility. To address this issue and mitigate permanent damage, an edema monitoring system was developed Yu et al., in which a dielectric electro-active polymer (DEAP) sensor detects capacitance changes in response to applied forces.

Other research teams investigated different microcirculation aspects, including capillary refill time (CRT). This parameter provides insight into peripheral arterial disease, including tissue breakdown and foot ulcerations, which can lead to more severe complications, such as amputation. The system developed in this study

incorporated an optical fiber sensor, as well as an FBG, to monitor blood volume change through Photoplethysmography (PPG), in addition to plantar pressure assessment [96].

In a study conducted by Bulathsinghala et al. [97], the authors proposed the development of a wearable sensory module capable of monitoring temperature variations as well as blood volumetric changes in the forefoot. Using acquired pulsatile blood flow signals, the authors were able to predict ischemic and neuropathic diabetic ulcerations. The proposed prototype also provided comprehensive monitoring by embedding the sensing module into a wearable bandage to provide real-time data that could be transmitted to a mobile application.

Expanding on these advancements in wearable technology for diabetic foot care, a recent research presented a sock designed to prevent foot inflammation in T2D patients [98]. These socks integrate sensors to monitor blood oxygen saturation, as well as temperature, to provide continuous noninvasive monitoring. Initial tests on healthy individuals and patients with early signs of inflammation demonstrated the system's ability to provide timely reminders through an application, emphasizing the potential to prevent diabetic foot complications. In parallel, another study explored the development of a smart shoe system designed to predict and prevent DFUs using Internet of Things (IoT) technology. This system features a flexible insole with integrated sensors to track oxygen saturation, along with plantar pressure,

Table 4 Research exploring biochemical parameters including microcirculation and pH as well as other attributes

Ref	Wearable	Measured Parameters	Num. of sensors	Clinically tested		Results
				Patients	Healthy	
[95]	DEAP stretch sensors	Edema				Edema was monitored using DEAP sensors.
[96]	Sensor unit	Blood volume	2 FBG sensors		✓	FBG sensors were used to detect blood volume changes, and the contact pressure applied to the foot.
[97]	Bandage	Blood volume and temperature		✓		The pulsatile blood flow sensor and temperature are integrated into a bandage.
[98]	Sock	Blood oxygen saturation and temperature	6 sensors	✓	✓	The designed sock is connected to an application that dynamically displays real-time readings of the measured parameters.
[99]	Insole	Oxygen saturation in blood, plantar pressure, temperature, and humidity	1 oxygen saturation sensor, 5 plantar pressure sensors, 5 temperature/humidity sensors.	✓	✓	The monitoring system presented accurate results with an accuracy of ($\pm 2.04\text{ N} \sim 0.20\text{ kg}$) for the plantar pressure sensor, ($\pm 0.24\text{ }^{\circ}\text{C}$) for the temperature sensors, ($\pm 2.14\%$) for the relative humidity, and (2%) for the oxygen saturation sensor.
[102]	Sensor gauze	pH and moisture				pH: 3 to 10 with a sensitivity of 0.01 pH/mv Moisture: 0.06%mv with a 0.1 response time.
[103]	Sensor unit	pH and temperature				pH: $\sim 42\text{ mV/pH}$ Temperature: High linearity between 25 $^{\circ}\text{C}$ and 50 $^{\circ}\text{C}$ with a sensitivity of $\sim 85\Omega/^{\circ}\text{C}$
[104]	Handheld sensors	pH and temperature		✓		pH and temperature were evaluated at the surface of wound.
[105]	Sock	Bioimpedance	3 sensors	✓	✓	Bioimpedance: 1–100 kHz
[106]	DIAPODAL sensor unit	Impedance plethysmography and plantar pressure	4 FSR sensors		✓	Using FSR sensors as plantar bioimpedance electrodes.
[107]	Sensor unit	Acceleration, rotation, galvanic skin response, Environmental temperature, humidity, force, skin temperature, and bioimpedance	42 sensors		✓	Accelerometer: $\pm 16\text{ g}$ Rotation: $\pm 2000\text{ s}^{-1}$ Humidity: 0 to 99%RH Temperature: -40 to 125 $^{\circ}\text{C}$ GSR: 0 to 5000k Ω Bioimpedance: 0 to 1023AU Force: 0 to 140 N Temperature skin: 20 to 40 $^{\circ}\text{C}$
[108]	Insole	Plantar strain and gait patterns	5 FBG sensors		✓	The FBG sensors in the array demonstrated an average temperature sensitivity ranging from 0.01637 to 0.02294 nm/ $^{\circ}\text{C}$, and a weight sensitivity ranging from 0.125 to 0.449 nm/g $\times 10^{-3}$

temperature and humidity. Upon detecting thresholds that may indicate the onset of complications, the system promptly alerts the user through the interface [99].

In a study by Armstrong et al. [100], an offloading device was developed to enhance blood circulation while simultaneously evaluating patient compliance. This work utilized the readily available OptiPulse device [101] to improve blood circulation and wound healing via intermittent plantar compressions which stimulate the pedal flow of oxygenated blood. These authors indicated the synergistic effect of plantar offloading, ensuring adequate flow of blood, and patient adherence is key to effective ulcer management.

pH level

Another chemical biomarker that exhibits a strong correlation with diabetic foot ulcers is pH. Changes in the skin's pH level can provide important information about the status of an ulcer contributing to its therapeutic management [73]. Furthermore, with diabetic foot ulcerations, the foot's temperature tends to increase, leading to excessive transpiration and hence to changes in the skin's pH level. Wound beds typically have an alkaline environment with a pH between 7.15 and 8.9. However, promoting a more acidic environment where the pH is between 4 and 7 has been reported to aid in wound healing by limiting bacterial infections of the wound, supporting

the proliferation of fibroblasts, and generating new blood vessels [72, 73].

Meanwhile, chronic wounds (CW), that do not exhibit proper/timely healing progress within four weeks, may signal an underlying bacterial infection contributing to the wound's chronicity. Accordingly, Du and Ciou [102], proposed a sensitized gauze designed to monitor the moisture and pH levels of a CW, providing the necessary protection while also assessing the healing progress. In a study by Slavo et al. [103], a temperature and pH measuring device was designed to monitor the DF's environment. The pH sensor included a graphene oxide layer in which the electrical potential changes in response to changes in the pH levels. Likewise, Mahindrakar et al. [104] studied the surface temperature and wound pH level in an effort to predict the healing of ulcers. The authors further confirmed that the microenvironment became increasingly more acidic as the ulcer progressively healed.

Associated attributes

Skin bioimpedance is another important parameter that provides input on tissue health by measuring its electrical properties. In the context of the diabetic foot, bioimpedance can offer insight into the hydration and composition of tissue, providing information about the skin and wound conditions [105]. Tronstad et al. [105] developed a system that can monitor tissue impedance using a smart customized sock. Morales et al. [106] also investigated the feasibility of using FSR sensors to measure bioimpedance along with pressure. The authors confirmed the successful evaluation using 4 FSR sensors.

Other researchers adopted a wider scope for monitoring multiple parameters reflecting the health of the diabetic foot by developing multimodal sensor platforms. One of these multimodal prototypes incorporated 42 sensors to gather data about several parameters, including temperature, humidity, impedance, as well as acceleration, rotation, and galvanic skin response in real-time [107]. Saskiumar et al. [109] proposed a diagnostic tool that uses Thermo Electric Generator (TEG) sensors to predict the DF where heat changes in the foot reflected decreased nervous system performance.

Mishra et al. developed and tested an advanced insole incorporating five Fiber Bragg Grating (FBG) sensors to measure plantar strain and analyze gait patterns [108]. Their aim was to address challenges in detecting diabetic foot ulcers (DFUs) by providing high spatial resolution and sensitivity. The study's findings highlight the effectiveness of closely spaced FBG arrays in capturing detailed strain distributions, underscoring their potential in early DFU detection and enhanced gait analysis. This innovative approach offers precise strain monitoring, making it a valuable tool in preventing diabetic foot

complications. Other studies also attempted to identify biomarkers capable of quantifying the chronicity of diabetic foot ulcers using electrochemical sensors [110].

Vigilant monitoring of various DF-relevant parameters is essential for devising effective wearable systems for preventing, monitoring, and managing diabetic foot ulcers. With rapid technological advancements, emerging innovations, and research efforts, multiple solutions have been introduced to further enhance the management of the diabetic foot. Vibrotactile actuation, in particular, provides a therapeutic approach that is promising in mitigating the impact of peripheral neuropathy on the diabetic foot.

Vibrotactile systems

The integration of vibrotactile technology into DF insoles offers potential benefits for pain management and sensory restoration. Studies show that whole-body vibrations (WBV) generated through electrical stimulation can inhibit the transmission of pain signals to the brain, modulate nervous system pathways, as well as improve the blood flow [111, 112]. Research also suggests that vibrotactile stimulation has the ability to enhance metabolic control by increasing the release of nitric oxide, hence promoting vasodilation [113, 114].

Sabziparvar et al. [112] studied the effectiveness of localized vibrations on the treatment of DPN. Employing a 62.5 Hz frequency for 10 min, they reported pain relief and an overall improvement in balance and sensory function after five treatment sessions, highlighting the potential of using this active technique as a treatment modality. Similarly, Gao et al. [115] evaluated a therapeutic shoe with a built-in vibration chip placed at the heel. Zhu et al. [113] also investigated the impact of applying local vibrations on the skin blood flow (SBF) by integrating a vibration chip into a wearable shoe. They reported that 10 min of stimulation at a 35 Hz frequency significantly increased the plantar SBF, indicating the possibility of using the vibrotactile system to reduce skin ischemia.

Furthermore, Hatton et al. [116] investigated the impact of vibrating insoles on the balance and muscle activity in patients with DPN. It was found that the vibrating insoles significantly reduced the center of pressure (CoP) sway and the ankle-foot muscle activity during challenging balance tasks. However, the insoles had an insignificant impact on gait patterns, suggesting that vibrotactile insoles primarily improve balance stability and neuromuscular control in DPN patients. In a related study, researchers explored the long-term effects of vibrating insoles on the vibration perception threshold (VPT). The study reported a significant improvement in the VPT after consistent use for 30 min daily over one month, demonstrating the potential to reduce the risk of diabetic foot ulcers by enhancing plantar sensation [114].

In a study conducted by Cham et al. [117] the impact of a vibro-medical insole was evaluated by investigating the pressure and vibration sensation in patients with DPN. The participants received customized insoles equipped with integrated vibratory systems. Substantial enhancement in pressure distribution and sensation were reported, suggesting the practical daily use of such insoles to overcome sensory loss.

Huchegowda et al. introduced an insole which measures the peak plantar pressure and displays the acquired information to a mobile application [118]. While a vibro-tactile system incorporated into the insole stimulates acupressure points in the foot to alleviate the pain. These authors highlighted the importance of using breathable materials in the development of footwear to maintain optimal temperature and humidity levels.

In general, several of the innovative smart solutions summarized in sect. “[Sensorized wearables](#)” be seamlessly integrated into smart wearables, offering practical modalities to continuously monitor DF-relevant parameters, relay the summarized information to a mobile application accessible by clinical teams/patients, as well as provide data-driven approaches to ulcer prevention and management options.

Unsensorized wearables

Researchers have also investigated alternative approaches to manage the diabetic foot, including passive solutions that employ smart materials and customized 3D printed designs. These interventions show promising potential in mitigating and controlling several DF-associated challenges, as described here.

Smart materials

Incorporating smart materials in the synthesis of footwear has a tremendous promising impact on preventing and reducing ulcers development. Suitable materials with characteristics related to pressure redistribution, stiffness, cushioning, breathability, absorption of moisture, and antibacterial properties, can be leveraged in wearables to protect the foot and create an optimal microenvironment.

In a study where shear-reducing insoles were compared to conventional footwear for high-risk DF patients, the authors reported that the novel insoles indicated great potential in reducing the risk of foot ulcerations [119]. Meanwhile, an alternative study explored the use of polypyrrole-coated commercial socks in controlling moisture absorption and bacterial effects, demonstrating the socks’ enhanced ability to absorb moisture and inhibit bacterial growth [120].

Zhou et al. [121] developed an intelligent fabric-based insole specifically for monitoring plantar pressure to prevent DFUs. The insole is constructed from wrap-knitted

spacer fabric, treated with a graphene composite coating, and combined with silicone rubber to form flexible sensors. The developed sensors exhibit a broad pressure response range and good sensitivity, enabling real-time monitoring. Furthermore, the insole was tested under high humidity conditions (90% humidity), confirming its robust pressure-sensing performance, thereby highlighting its potential as a reliable tool for continuous monitoring in various environments.

Jafarzadeh et al. [122] explored the use of orthopedic insoles and their efficacy in pressure distribution, specifically focusing on the impact of the shapes and stiffness of various insoles on plantar pressure reduction. The optimized design showed a 40% and 25% reduction in the maximum pressure when the optimal stiffness and shape were employed, respectively. Likewise, Nouman et al. [123] studied pad design and material stiffness and their impact on the plantar pressure using FEA, confirming the importance of appropriate design and material selection. In a different investigation led by Shi et al. [24], contoured insoles utilizing different materials were quantitatively compared to analyze their effect on gait. The researchers reported the positive impact of 3D contoured insoles towards preventing ulcerations. Additionally, Maharana et al. [124] proposed a unique dynamic wearable, designed to mechanically reduce high plantar pressure. The 3D-printed prototype encompassed an array of snapping arches actuated by the body’s weight.

In an alternative perspective adopted by Chatzistergos et al. [125], footwear cushioning stiffness was investigated. The authors reported the benefits of employing optimum cushioning to reduce pressure. This study indicated that since different body masses require varied material stiffness, 3D printing technology can be used to custom tailor wearables to the prescribed mechanical properties.

3D printed wearables

Multiple studies have utilized 3D printing technology to create wearables characterized by their ability to reduce plantar pressure and improve the overall health of the feet and lower limbs. Using this approach, insoles and other types of footwear can be customized to meet the required patient-specific optimal support needs while selecting suitable materials. A comfortable fit of the wearable can enhance the adherence of patients, thereby promoting effective pressure reduction.

In a study conducted by Ahmed et al. [126], clinical footwear was studied to identify optimal design features that effectively offload plantar pressure and enhance the adherence of diabetic users. Beach et al. [127] also developed a 3D-printed wearable system incorporating 4 temperature sensors strategically placed at different ROIs. The system rendered relevant information about

temperature variations across different regions in the foot, hence providing insight into biomarkers related to the biomechanics and vascularization of different soft tissues in the course of disease progression. Rapid advancements in custom-made wearables are continuously evolving to encompass specific microstructures within materials while also incorporating hybrid blends in fabrication to further mitigate DF complications [128]. For instance, Leung et al. [129] constructed a heel pad featuring an innovative honeycomb structure which helped alleviate pressure on the heel by increasing the contact area and sufficiently deforming to accommodate the heel. These authors reported a significant decrease in the maximum force and average pressure of 33% and 17%, respectively. Other research teams integrated 3D printing technology with custom-made insoles which included smart sensing modalities to monitor specific parameters [130].

Shaulian et al. [131] explored an offloading heel support model to better distribute the plantar pressure. Employing a Pedar pressure measuring system, this study revealed that using a heel pad with a hollow center beneath the ulcerated tissue can lead to elevated pressure on the periphery of the offloaded region. A custom 3D printed pad configuration was consequently devised to address this issue and reduce the pressure.

AI technology and the diabetic foot

In response to the continued challenges in dealing with diabetic foot and its wide spectrum of complications, AI has recently emerged as a promising technology for addressing various aspects of diabetic foot care. Clinically, AI has the potential to revolutionize diabetic foot care by enhancing the accuracy and efficiency of diagnosis enabling personalized treatment plans and integrating seamlessly with traditional practices to improve patient outcomes/. One significant application of AI is the timely screening and detection of foot ulcers, which is a critical step in avoiding serious complications such as amputation [132]. AI techniques, when integrated with proper sensor technology, have the potential to detect foot ulcers early, allowing for timely intervention. Several studies have demonstrated the application of AI in this context, combining data from various sources to deliver promising results [133, 134].

For instance, Cassidy et al. [133] evaluated an AI system integrated into a smartphone for automated detection of diabetic foot ulcers. The system demonstrated excellent sensitivity and high specificity as compared to expert clinician judgment, highlighting its potential as a valuable diagnostic tool. Moreover, it showed high levels of inter- and intra-rater reliability, indicating consistent performance across different raters.

Another area where AI technology has shown potential is remote monitoring of diabetic foot ulcers. Traditionally, evaluating the progression of foot ulcers involved frequent clinic visits, which presented practical issues for both patients and healthcare personnel and negatively impacted patient adherence. However, the introduction of AI-powered mobile apps allows patients to conveniently take DF/ulcer photographs with their normal cell phones and send them to healthcare specialists via remote monitoring. This not only increases access to care but also enables more frequent and prompt examinations, resulting in improved foot ulcer management [135]. Furthermore, AI-based technologies have the potential to transform wound care by providing objective and quantitative measures of ulcer healing progress [134, 136]. By assessing criteria such as ulcer size, depth, surrounding tissue features and mechanical properties, AI systems can assist clinicians in making informed data-driven treatment recommendations. In a related study by Zhang et al. [137], a novel approach was introduced to predict dynamic plantar pressure in diabetic patients using ink footprint images and a multilayer perceptron (MLP) model with localization embedding. The researchers developed a dataset by collecting and analyzing footprint images and pressure data for diabetic patients, which was then used to train the model. This model outperformed traditional methods in terms of efficiency and accuracy, offering a promising tool for clinicians to recommend patient-specific insole materials and designs, potentially aiding in the prevention of diabetic foot ulcers.

Despite technological advances, there remain many challenges to overcome in integrating AI technology into diabetic foot care. The heterogeneity of data sources, as well as the variability in machine learning algorithms utilized in different studies, currently present constraints to standardization and scalability. The integration of sensing modalities and smart wearables with novel actuation technology and AI tools can enable real-time prevention, management, and effective wound healing of the diabetic foot, offering personalized and proactive care approaches that promise to introduce a paradigm shift in current diabetic foot management.

Limitations

Wearable technology designed to monitor and manage diabetic foot health has made notable progress, according to this review, yet it continues to contend with several limitations. Accuracy remains a primary concern, particularly in detecting early signs of diabetic foot complications such as neuropathy or ulcer formation. While most of the reviewed devices are able to track surface-level measurements, such as steps, peak pressure, or skin temperature, they often lack the capability to detect

deeper physiological changes that indicate underlying foot problems, such as microcirculation, ischemia, pH balance and wound healing attributes. Comfort and wearability are additional challenges, as some wearables may prove bulky for extended use, potentially discouraging patient adherence. Moreover, interpreting the vast amounts of data generated by these devices can be daunting for users, hindering effective foot health management. Integration with healthcare systems poses another hurdle, with many wearables operating independently of electronic health records or healthcare provider systems. Cost, size, weight, privacy, and customization concerns further underscore the need for further advancements in sensor and actuator technology, 3D printing, smart materials, computational tools and data analytics, and user interface designs towards enhancing the efficacy, accessibility, and adherence level of smart wearable solutions for diabetic foot care.

Future recommendations

The articles considered in this review highlight the importance of harnessing current and emerging smart sensing modalities capable of capturing diverse complications associated with DF health. This requires a multidisciplinary thorough exploration of both the physical/mechanical and chemical/clinical parameters influencing foot health, as well as computational technologies and AI capabilities for synergistically and simultaneously monitoring, classifying, and predicting all the key parameters. The developed prototype should be validated by rigorous laboratory and clinical trials to analyze its functionality towards timely prediction/mitigation of foot complications. It is equally important to explore performance and effectiveness in different settings and environments, encompassing both indoor and outdoor milieus to evaluate the robustness and overall comfort of the design. It is also valuable to address patient, gender, and culture-specific habits and needs, to ensure effectiveness and patient compliance. Moreover, when developing such a device, it is important to consider the utilization of energy-efficient technologies rendering wearables that are sustainable and cost-effective. Other crucial aspects include investigating the long-term effects and the functionality of the wearable while also taking into consideration the safety of such a device as well as user satisfaction. It is also practical to create modular designs that address the changes of the foot where components can be adjusted to accommodate or replace specific parts to extend the lifespan of the footwear.

The incorporation of cost-effective flexible and thin printable sensors that are seamlessly integrated into wearables without bulky designs can encourage users to adhere to therapeutic footwear. Accordingly, all acquired sensor readings can be presented through smart device

applications integrated with AI to analyze data, enabling classification and predictive analytics as well as early detection of anomalies. The realm of remote monitoring allows for quicker interventions which can be easily conducted through telemedicine, thereby reducing the necessity for frequent hospital visits and enhancing patient compliance. Importantly, strategic integration of key features such as personalized engagement, alerts, and reminders into the system can greatly impact the trajectory of loading and ulcer healing, especially in age, gender, and culturally specific contexts. Furthermore, the addition of other active features, including vibrotactile actuators, electrical muscle stimulation, and red-light therapy may aid in continuous timely management and pain alleviation. Overall, the adoption of a user-centered design (UCD) approach, by engaging the various stakeholders (patients, caregivers, clinicians, engineers, etc.) through the design and development journey, allows for patient-specific tailored innovation, while leveraging the current state-of-the-art technology towards custom-designed smart footwear with multimodal sensing and actuation capabilities. Such wearables, along with patient/population-specific applications and AI-based predictive analytic capabilities, can greatly prevent limb loss and enhance the quality of life of T2D patients by protecting them from preventable complications that hinder independence and day-to-day activities and instigate huge socioeconomic challenges.

Conclusion

This review highlights the latest developments in wearable technology for the prevention and management of diabetic foot complications covering both active and passive strategies. We evaluated the use of various sensing modalities, including plantar pressure, shear loading, temperature, humidity, pH, microcirculation, and skin impedance, and their crucial roles in predicting and managing DF.

The integration of wearable technology into clinical practice offers several benefits including continuous monitoring and real-time biofeedback. This allows the detection of early changes in DF skin barrier health that could progress to ulceration and infection enhancing current foot screening practices by enabling timely interventions before minor issues escalate into serious limb-threatening complications. It is important to recognize that wearable technology is designed to support, not replace, the expertise of trained clinicians and regular diabetic foot assessments. The use of smart materials and 3D printing in the design of wearables further improves their comfort, adherence, and effectiveness. Custom-tailored wearables that meet individual patient needs can also significantly enhance clinical outcomes.

Future research should aim to integrate these advanced wearables with existing healthcare systems to streamline data collection, analysis, and clinical decision-making. This integration will enhance the overall standard of care for patients with diabetes, making a meaningful impact on their health and quality of life.

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Author contributions

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

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